

Lab 4

- Motor Schemas implemented are phototaxis, obstacle avoidance and random walk
- The perceptual schema of phototaxis:
 - Read all light sensors
 - Find the sensor with the highest light intensity
 - Use its angle as the direction of movement
- Vector used for phototaxis:
 - Angle = direction of the stronger light
 - Length = proportional to the light intensity
- The perceptual schema of obstacle avoidance schema
 - Read all proximity sensors
 - Find the sensor with the highest proximity value
- The vector used for obstacle avoidance
 - Angle = rotated 90 degrees from the obstacle direction
 - Length = proportional to the proximity value
- Perceptual schema of random walk
 - Activated only if the light sensors do not detect any light
- Vector of random walk
 - Both speed and angle are random
- Purpose of light threshold
 - To keep the robot near the light source once it reaches the light threshold was introduced.
 - When the light sensor detects an intensity greater than or equal to the threshold value, the controller considers the target reached and stops generating the vectors.
 - As a result the robot remains close to the light source and on top of the black spot
 - An alternative solution was to use the ground sensors to detect the black spot located under the light source (as implemented in previous labs).
 - To keep code clean and readable I chose a light threshold approach.